

PROBABILITY THEORY AND RANDOM PROCESSES (MA225)

LECTURE SLIDES

Lecture 30

Positive and Null Recurrent

Def: A recurrent state i is said to be positive recurrent if

$$m_i = E_i(T_i) < \infty.$$

If i is not positive recurrent, then it is called null recurrent.

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible MC. Then the following are equivalent.

- ① Every state is positive recurrent.
- ② Some state i is positive recurrent.
- ③ The MC has a stationary distribution π .

Moreover, when (3) holds, $\pi_i = \frac{1}{m_i}$ for all $i \in S$.

Remark: Positive recurrence and null recurrence are class properties.

Example

Example 1: Recall SRW. We have proved that asymmetric simple random walk is transient. Thus, by previous theorem, it cannot have a stationary distribution.

Example 2: For SSRW, we have shown that all states are recurrent. Now, $\pi_i = 1$ for all i satisfy $\pi_i = \frac{1}{2}\pi_{i-1} + \frac{1}{2}\pi_{i+1}$. Hence, π is an invariant measure. Since, any invariant measure must be a scalar multiple of π , and $\sum_i \pi_i = \infty$, there cannot exist a stationary distribution. This also shows that all states are null recurrent.

Proof of the Theorem

Theorem: Let $\{X_n : n \geq 0\}$ be an irreducible MC. Then the following are equivalent.

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Moreover, when (3) holds, $\pi_i = \frac{1}{m_i}$ for all $i \in S$.

Proof: (1) $\implies$ (2) is obvious.

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Proof: (2) $\implies$ (3).

- If i is positive recurrent, it is definitely recurrent. So, by irreducibility, the chain is recurrent.
- Thus, γ^i is an invariant measure.
- But, $\sum_{j \in S} \gamma_j^i = m_i < \infty$.
- Hence, $(\pi_j = \gamma_j^i / m_i : j \in S)$ gives an invariant distribution.

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Proof: (3) $\implies$ (1).

- Consider any state k .
- Since, $\sum_{i \in S} \pi_i = 1$, there exists an i such that $\pi_i > 0$.
- Now, by irreducibility, there exists n with $p_{ik}^{(n)} > 0$. Thus,
$$\pi_k = \sum_{j \in S} \pi_j p_{jk}^{(n)} > 0.$$
- Set $\lambda_i = \frac{\pi_i}{\pi_k}$. Then λ is an invariant measure with $\lambda_k = 1$.
- So, by previous theorem, $\lambda \geq \gamma^k$.

Proof of the Theorem

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- ① Every state is positive recurrent.
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Moreover, when (3) holds, $\pi_i = \frac{1}{m_i}$ for all $i \in S$.

Proof: (3) $\implies$ (1).

- Hence,

$$m_k = \sum_{i \in S} \gamma_i^k \leq \sum_{i \in S} \frac{\pi_i}{\pi_k} = \frac{1}{\pi_k} < \infty.$$

- Thus, k is positive recurrent.
- Now, if the chain is recurrent, we have $\lambda = \gamma^k$ and hence

$$m_k = \sum_{i \in S} \gamma_i^k = \sum_{i \in S} \frac{\pi_i}{\pi_k} = \frac{1}{\pi_k}.$$

Example

Example 3: Consider the MC with state space $S = \{0, 1, 2, \dots\}$ and transition probabilities

$$p_{0,1} = 1, \quad p_{i,i+1} = p_{i,0} = \frac{1}{2} \quad \text{for all } i \geq 1.$$

- It is easy to see that the MC is irreducible.
- Let $\pi = (\pi_i : i \in S)$ be a stationary distribution, then

$$\pi_0 = \frac{1}{2} (\pi_1 + \pi_2 + \dots),$$

$$\pi_1 = \pi_0, \pi_2 = \frac{\pi_1}{2} = \frac{\pi_0}{2}, \pi_3 = \frac{\pi_2}{2} = \frac{\pi_0}{2^2}, \dots, \pi_n = \frac{\pi_0}{2^{n-1}}, \dots$$

- As $\sum_{i=0}^{\infty} \pi_i = 1$,

$$\pi_0 + \pi_0 + \frac{1}{2}\pi_0 + \frac{1}{2^2}\pi_0 + \dots = 1 \implies \pi_0 (1 + 2) = 1 \implies \pi_0 = \frac{1}{3}.$$

- Thus, the unique stationary distribution is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3 \cdot 2}, \frac{1}{3 \cdot 2^2}, \dots)$.